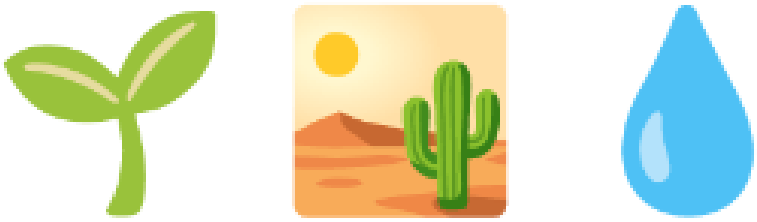
🌍 Warming Up: Dark vs. Light Surfaces 🌍

Physical Science — Sunshine & Warmth — Unit 2, Lesson 3 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth
🔢 **Lesson:** 3 of 6



Soil, sand, and water — three surfaces, three different ways of warming up!

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate warming Up: Dark vs. Light Surfaces, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Earth has many different surfaces — soil, grass, sand, pavement, water, ice. Each one warms up at a **different rate** when sunlight hits it. Dark surfaces like soil absorb more heat and warm quickly. Light-colored surfaces like sand warm more slowly. Water warms the slowest of all! Understanding this helps us understand weather, climate, and even why cities can feel much hotter than the countryside.

🔬 The Science

Different materials have different **heat capacities** — how much energy it takes to warm them up. Water has a very high heat capacity, meaning it needs a lot of energy to increase in temperature. That's why the ocean stays cool even on hot summer days. Dark soil absorbs light efficiently and heats up quickly. Sand is lighter in color and also has lower heat capacity than water, so it warms faster than water but often slower than dark soil. This explains the **urban heat island effect**: cities filled with dark pavement and rooftops absorb heat all day and stay hotter at night than nearby green areas.

> 💡 **Key vocabulary:** surface, absorb, reflect, soil, sand, pavement, urban heat island, temperature, compare

📦 Materials

🏠 What You Need

- Dark soil (potting mix works great)
- Light sand or flour (as a light-surface substitute)
- Water
- 3 clear plastic cups or small bowls
- A sunny spot (windowsill, porch, or backyard)
- Timer (15 minutes)
- Thermometer (optional but exciting!)
- Student worksheet (included)

🔬 Let's Get Started

Step 1 — Set Up the Cups (5 min)
Label 3 cups: "Soil," "Sand," and "Water."
Fill each about halfway. Note the colors: soil is dark, sand/flour is light, water is nearly clear.
Ask: *"Which do you think will feel warmest after sitting in the sun? Why?"*
Have students record their predictions on the worksheet.

Step 2 — Into the Sun! (15 min)
Place all three cups in the same sunny spot — same position, same sunlight exposure.
Set a timer for 15 minutes. While waiting, discuss: *"Have you ever noticed that pavement feels really hot on a summer day, but grass feels cooler? Why might that be?"*

Step 3 — Feel & Compare (5 min)
After 15 minutes, carefully feel the surface of each cup's contents (touch the top of the soil, sand, or water).
If using a thermometer, measure each cup and record the numbers!
Ask: *"Which is warmest? Which is coolest? Were your predictions right?"*

Step 4 — Connect to the Real World (10 min)
Talk about Earth's surfaces: *"Our cups are like tiny versions of Earth's surfaces. What does the soil cup represent? What about the water cup?"*
Guide them to Earth's soil, deserts, oceans, and paved cities.
Introduce the idea of the urban heat island: cities with lots of pavement and dark roofs get much hotter than areas with grass and trees.

💬 Discussion Question

🗣️ **"Why do cities feel hotter than the countryside on sunny days?"**

> *→ Cities have lots of dark pavement and rooftops that absorb sunlight and hold onto that heat. The countryside has grass, trees, and water that reflect more light and cool down through evaporation. This difference is called the "urban heat island effect" — cities can be 5–10 degrees warmer than nearby rural areas on the same day!*

🚀 Extensions

- 🌡️ **Temperature Race:** If you have a thermometer, check the temperature every 5 minutes and make a simple graph — or just note warmest/coolest at end.
- 🏙️ **City vs. Country:** Look up "urban heat island" images online together. The satellite thermal photos showing cities glowing orange-red compared to surrounding green countryside are striking!
- 🌊 **Cooling Effect:** On a hot day, stand on pavement barefoot vs. standing in grass. Which feels cooler? Now you know why!
- 🌴 **Green Solution:** Plant a tree, add grass, or put out a tray of water near a thermometer. Watch temperatures drop! Cities are now planting millions of trees to fight urban heat islands.

👩‍🏫 Parent/Teacher Notes

- **Flour vs. sand:** If you don't have sand, white flour works well as a light-surface substitute. It behaves similarly and is usually on hand.
- **Water results:** Water will feel noticeably cooler than the other two after 15 minutes. If using a thermometer, this difference is very measurable and impressive to kids.
- **Real-world connection:** If your family has visited the beach, remind kids how the sand gets scorching hot while the ocean stays cool — a perfect real-life example from Lesson 3.
- **Building on Lesson 1:** In Lesson 1 they found dark paper warms fastest. This lesson extends that — now they're comparing natural Earth materials, not just paper colors.
- **Assessment:** Can the child rank the three surfaces? Can they connect it to a real-world example (sand on a beach, a city parking lot, a lake)?

🌍 Student Worksheet

Name: _____ Date: _____

🔮 My Predictions — Circle your prediction for each cup:

🌱

Dark Soil

After 15 min, this will be:

❄️ **Cool** 😐 **Warm** 🔥 **Hot**

circle one

🏜️

Light Sand

After 15 min, this will be:

❄️ **Cool** 😐 **Warm** 🔥 **Hot**

circle one

💧

Water

After 15 min, this will be:

❄️ **Cool** 😐 **Warm** 🔥 **Hot**

circle one

👀 What I Observed — Fill in the blanks:

🌱

Soil felt...

_____

write or draw

🏜️

Sand felt...

_____

write or draw

💧

Water felt...

_____

write or draw

🏆 Rank the surfaces from WARMEST to COOLEST:

🥇 Warmest	🥈 Middle	🥉 Coolest

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Expected Warmth Ranking (after 15 min in sun)

1. 🔥 **Dark soil** — warmest. Dark color absorbs light; soil also has lower heat capacity than water
2. 😐 **Sand/flour** — middle. Lighter color reflects some light; also has moderate heat capacity
3. ❄️ **Water** — coolest. Water has very high heat capacity, requires a lot of energy to warm up

Note: With a thermometer, you may see soil 10–15°F warmer than water after 15 min in direct sunlight.

💡 Science Explanation for Parents

Heat capacity is the amount of energy required to raise the temperature of a material. Water has an extremely high heat capacity — one of the highest of any common substance. This means it takes a lot of energy to warm it up, but also that it holds heat for a long time once warm. Soil and sand have lower heat capacities and darker colors, so they heat up quickly but also cool down faster. This is why ocean temperatures are much more stable than land temperatures, and why coastal cities have milder climates than inland cities.

The **urban heat island effect** is real and measurable. Cities can be 5–10°F (3–6°C) warmer than nearby rural areas on the same day, due to dark paved surfaces, waste heat from buildings and vehicles, and reduced vegetation (which would otherwise provide cooling through evapotranspiration).

✅ What Success Looks Like

- Child can correctly rank the three materials (even approximately)
- Child can connect dark soil to dark pavement / city heat
- Child can name a real-world example: "The beach sand gets hot but the ocean is cool"

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